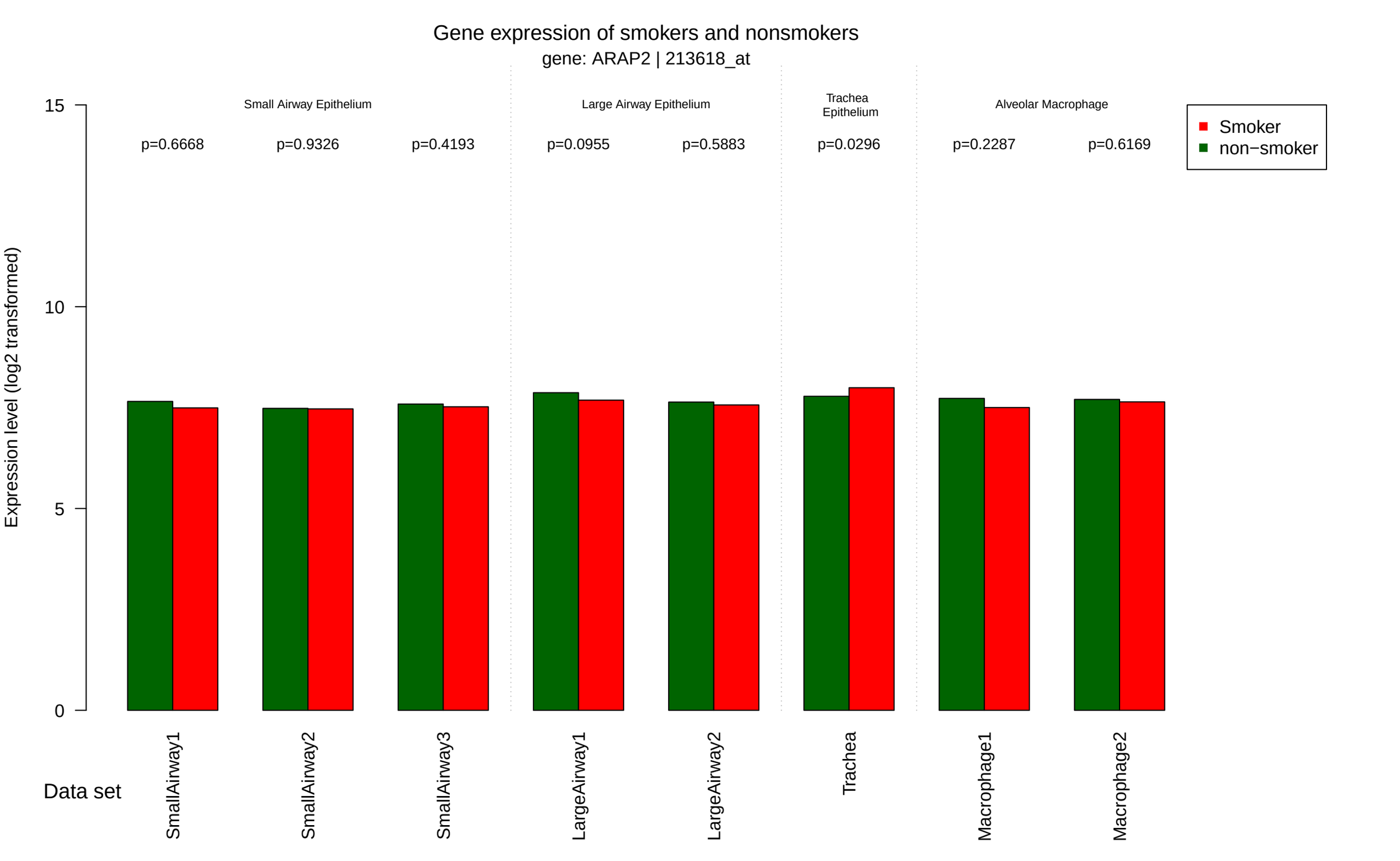
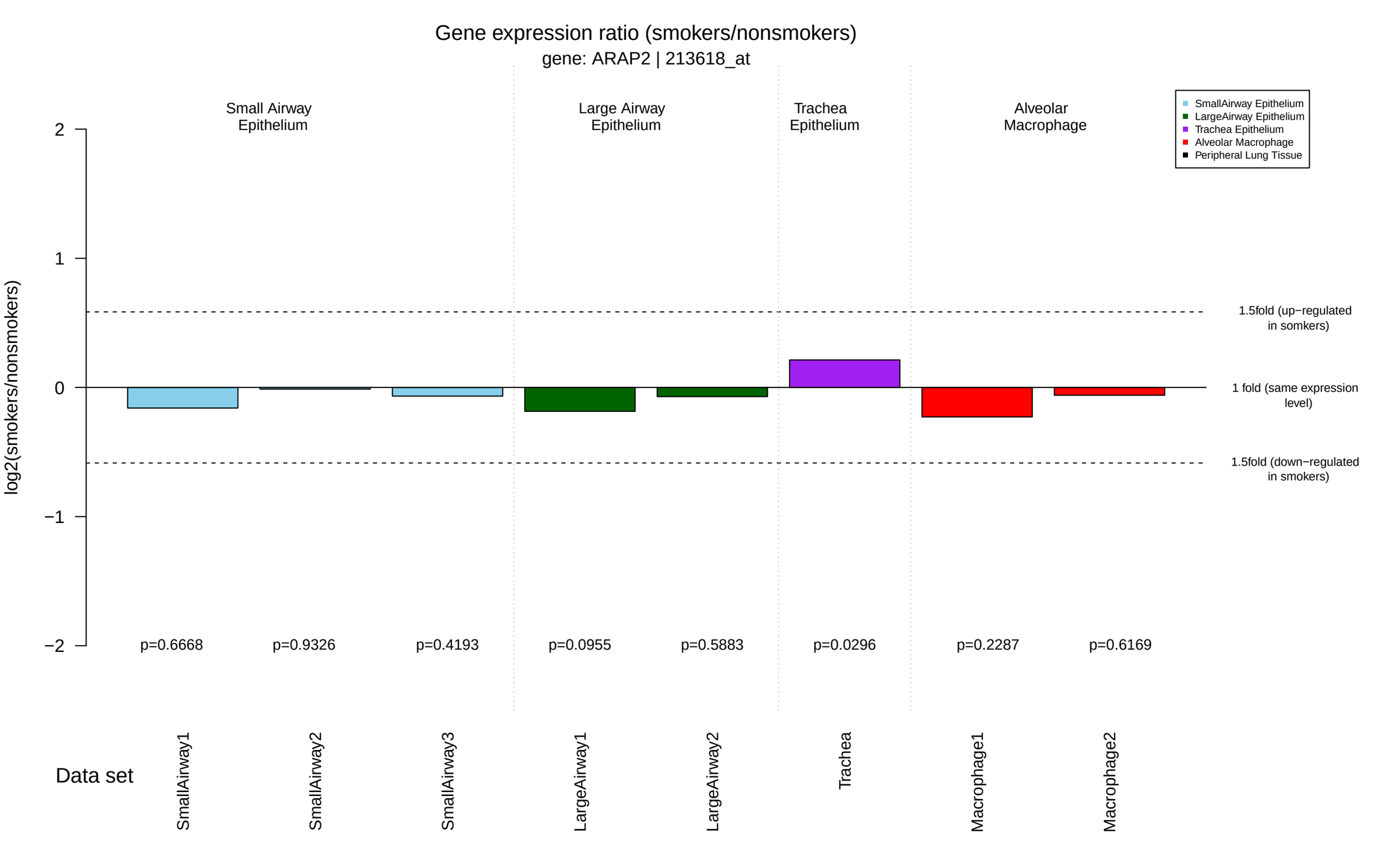
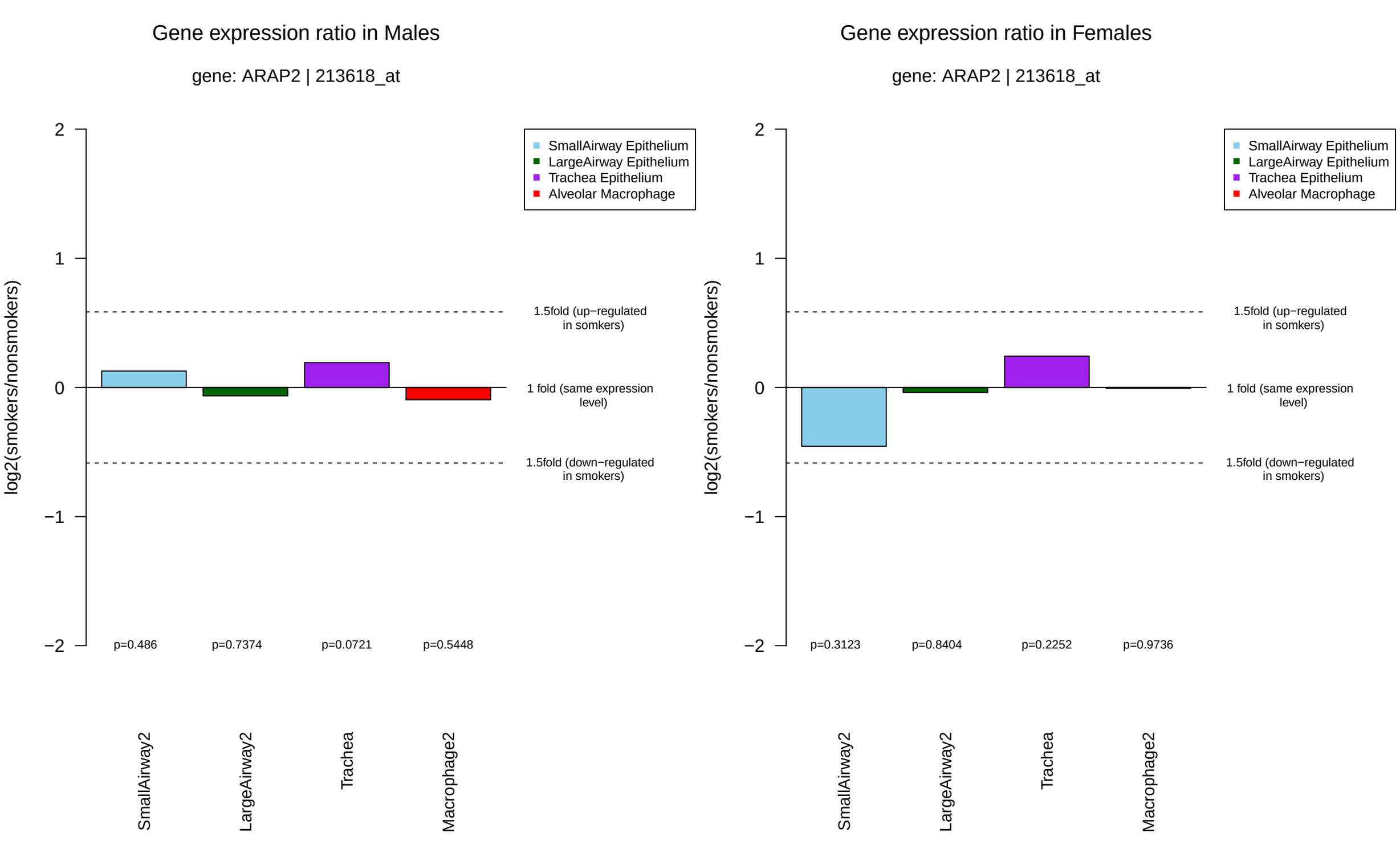
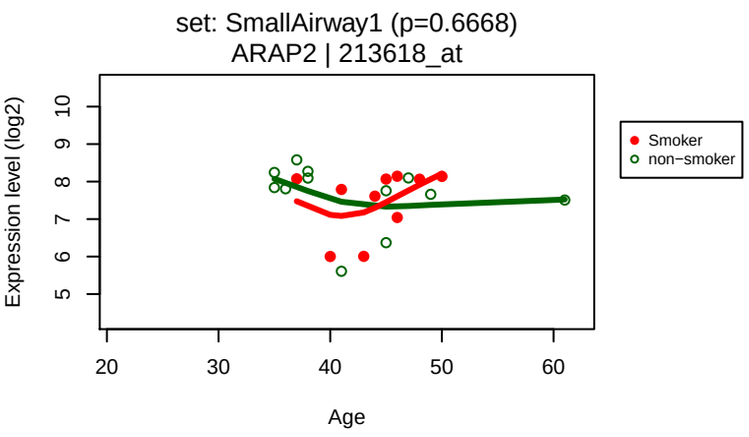
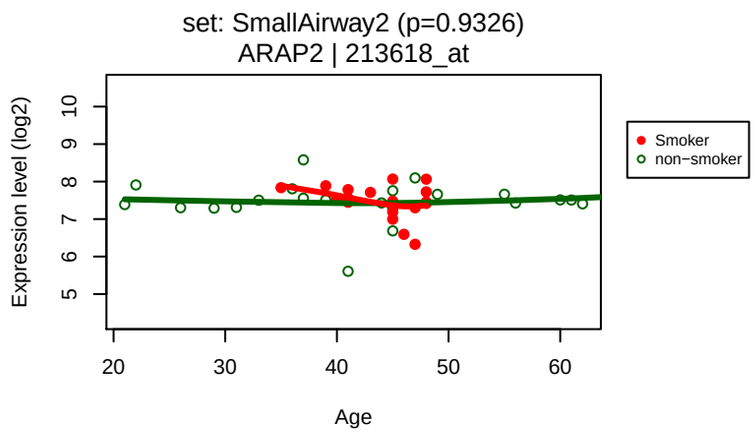
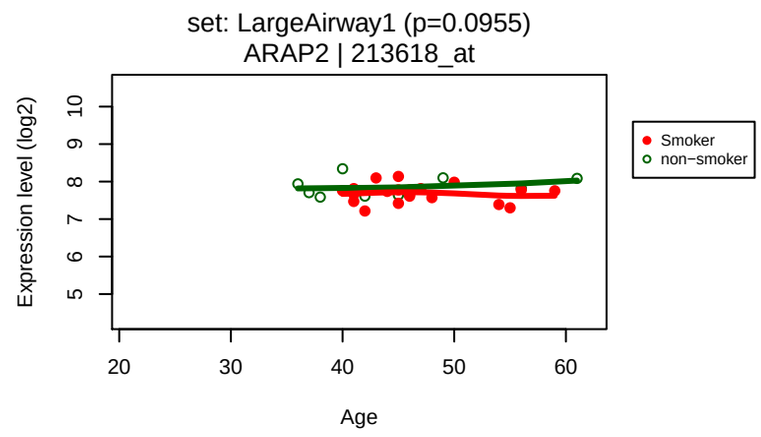

Gene expression of smokers and nonsmokers
gene: ARAP2 | 213618_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6668
p=0.9326
p=0.4193
p=0.0955
p=0.5883
p=0.0296
p=0.2287
p=0.6169
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

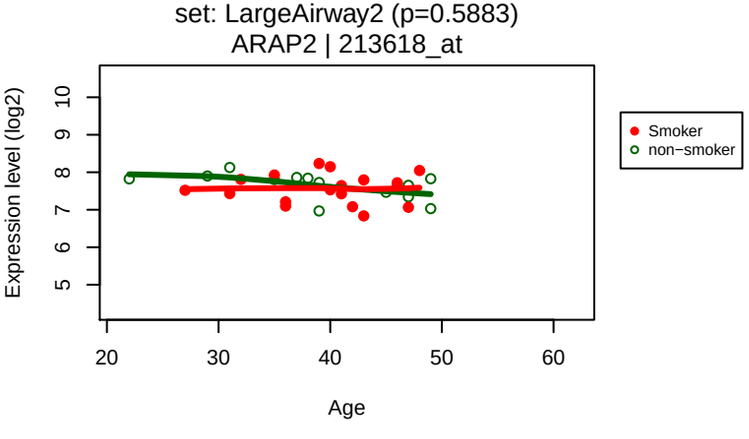

Gene expression ratio (smokers/nonsmokers)
gene: ARAP2 | 213618_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6668
p=0.9326
p=0.4193
p=0.0955
p=0.5883
p=0.0296
p=0.2287
p=0.6169
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

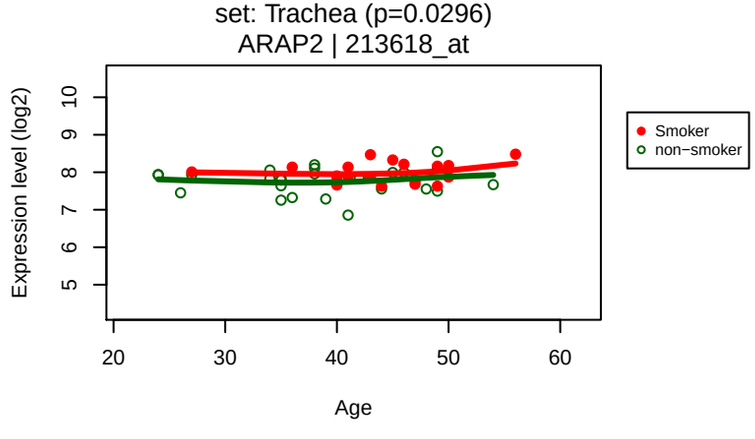

Gene expression ratio in Males
gene: ARAP2 | 213618_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.486
p=0.7374
p=0.0721
p=0.5448
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARAP2 | 213618_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3123
p=0.8404
p=0.2252
p=0.9736
SmallAirway2
LargeAirway2
Trachea
Macrophage2

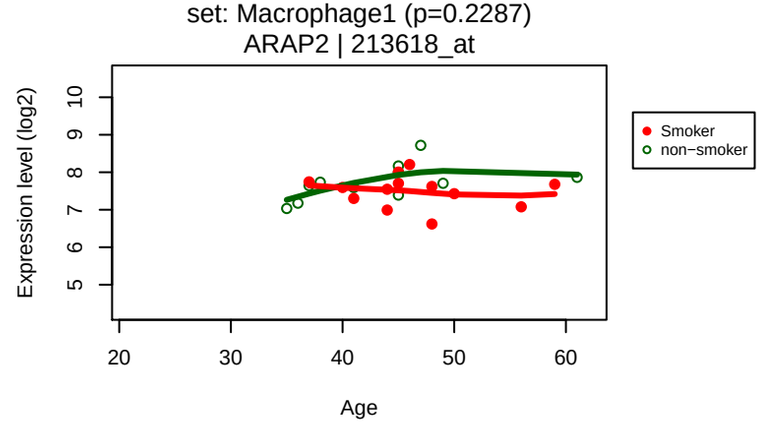

set: SmallAirway1 (p=0.6668)
ARAP2 | 213618_at
Expression level (log2)
Age
Smoker
non−smoker

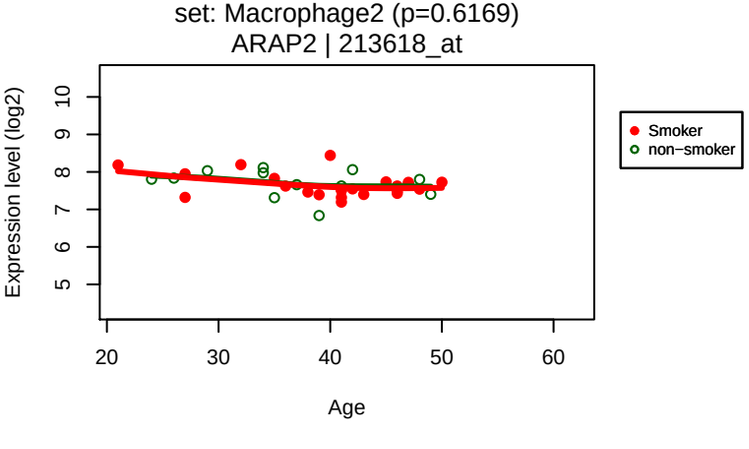

set: SmallAirway2 (p=0.9326)
ARAP2 | 213618_at
Expression level (log2)
Age
Smoker
non−smoker

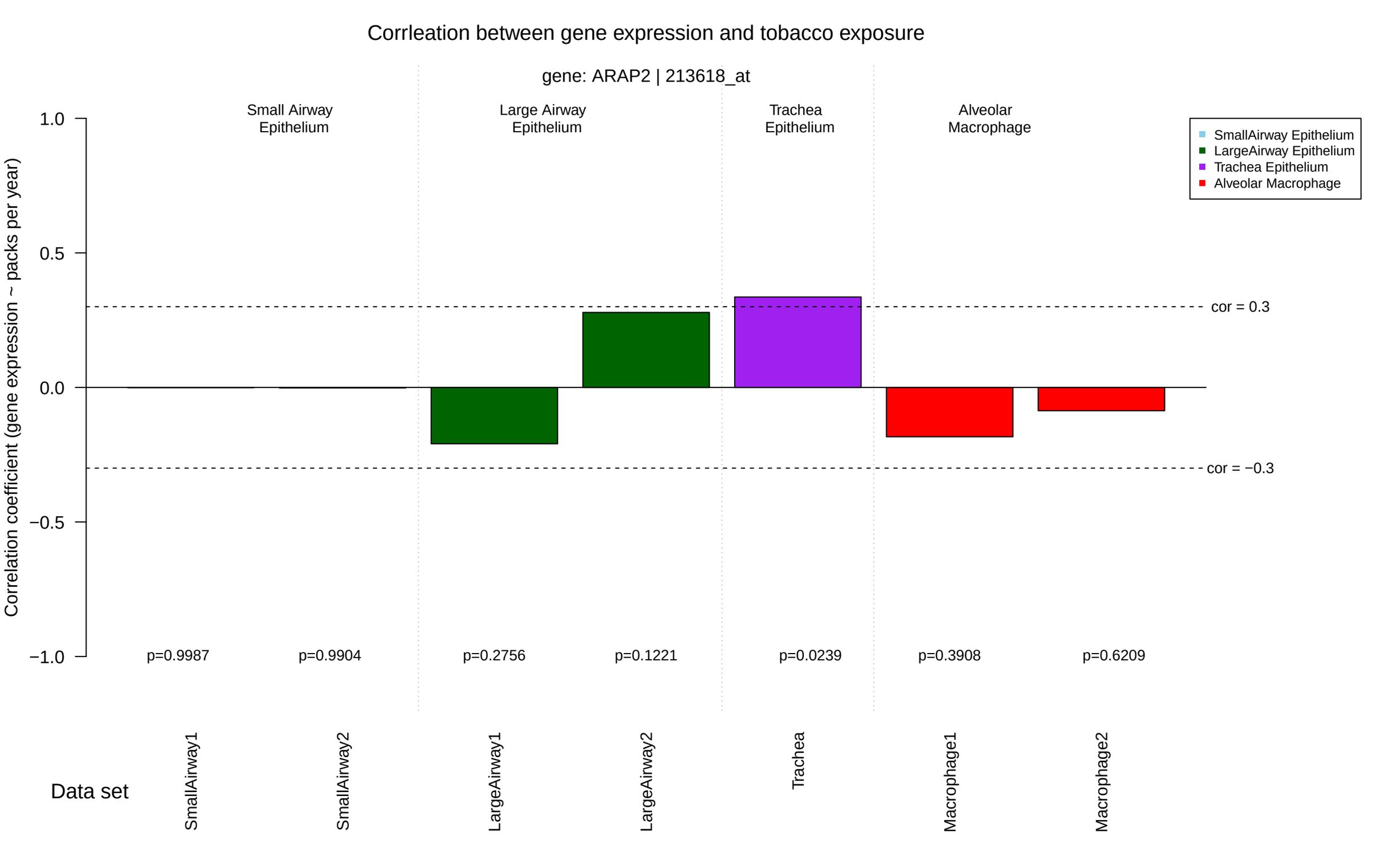

set: LargeAirway1 (p=0.0955)
ARAP2 | 213618_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5883)
ARAP2 | 213618_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0296)
ARAP2 | 213618_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2287)
ARAP2 | 213618_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6169)
ARAP2 | 213618_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARAP2 | 213618_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9987
p=0.9904
p=0.2756
p=0.1221
p=0.0239
p=0.3908
p=0.6209
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2